Comparing Pretrained Arabic Language Models on Intended-Meaning Classification of Saudi Dialect Text

Course project (Deep Learning). We build a four-class Saudi-dialect dataset (**Positive, Negative, Neutral, Ironic**) from three sources, label 1,200 texts by hand, and compare four pretrained Arabic encoders (AraBERT, MARBERT, MARBERTv2, SaudiBERT) with stratified 5-fold cross-validation and leave-one-source-out (LOSO) testing.

Repository structure

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├── README.md
├── requirements.txt
├── notebooks/
│   ├── 01_data_preparation.ipynb        # clean, merge, describe, draw the sampl
│   └── 02_5fold_cv_four_models.ipynb    # baselines, 5-fold CV, LOSO, pseudo-lab
├── data/
│   ├── README.md                        # sources, licenses, how to obtain the r
│   ├── merged_clean.csv                 # cleaned and merged texts (only if the
│   └── manual_gold.csv                  # 1,200 manually labeled texts (id, sour
├── results/                             # all CSV/JSON outputs of notebook 02 (a
├── figures/                             # result figures and the data and method
└── docs/                                # induction proof and paper PDF
```

Data

Source	Original labels	Reference
Sa'7r	ironic / not ironic	AlMazrua et al., OSACT 2022
Telecom (AraCust)	positive / negative	Almuqren and Cristea, PeerJ CS 2021

Bank (Saudi Bank Sentiment Dataset, 12,048 tweets about four Saudi banks)	positive / negative / neutral	Alqahtani et al., ICNLSP 2022; https://github.com/iwan-rg/Saudi-Bank-Sentiment-Dataset (check its license)

Raw datasets are not redistributed unless their licenses allow it. See `data/README.md` for where to download them.

Gold set (manual labels, 1,200 texts): annotator 1 labeled all rows, annotator 2 labeled 200 (Cohen's kappa = 0.70).

Source	Positive	Negative	Neutral	Ironic
Bank	61	225	140	46
Sa'7r	60	68	42	238
Telecom	92	145	39	44
Total	**213**	**438**	**221**	**328**

Label definitions: **Positive** = compliment or satisfaction; **Negative** = complaint; **Ironic** = surface wording opposite to the intended meaning; **Neutral** = no evaluative or hidden meaning.

How to reproduce (Google Colab, GPU)

1. Download the raw datasets (see `data/README.md`).
2. Open `notebooks/01_data_preparation.ipynb`, upload the three files, run all cells. It produces `merged_clean.csv`, `cleaning_log.csv`, and the file to label by hand.
3. Label the `label` column manually (Positive, Negative, Neutral, Ironic) and save it as `labeled.csv` (or use the provided `data/manual_gold.csv`).
4. Open `notebooks/02_5fold_cv_four_models.ipynb`, upload `manual_gold.csv` and `merged_clean.csv`, and run all cells. Set `USE_DRIVE = True` to keep results across disconnects. Approximate time on a T4 GPU: 30 to 40 min (CV), 15 min (LOSO), 25 min (ablation).

Main results (stratified 5-fold CV on the 1,200 gold texts)

			95% bootstrap CI (OOF

Model	Accuracy	Macro-F1	Macro-F1)
Majority class	0.365 ± 0.002	0.134 ± 0.001	[0.126, 0.141]
TF-IDF + Logistic Regression	0.692 ± 0.016	0.666 ± 0.016	[0.642, 0.695]
AraBERT	0.784 ± 0.025	0.771 ± 0.027	[0.747, 0.797]
MARBERT	0.791 ± 0.024	0.775 ± 0.023	[0.749, 0.800]
MARBERTv2	0.771 ± 0.033	0.755 ± 0.035	[0.731, 0.780]
SaudiBERT	0.802 ± 0.030	0.787 ± 0.031	[0.761, 0.811]

The confidence intervals of the four transformers overlap, so differences between them are small. LOSO results, per-class and per-source scores, and the pseudo-labeling ablation (no improvement) are in `results/`.

Reproducibility details

- Random seed 42 (fold seed = 42 + fold index); fold assignments are saved in `results/fold_assignments.csv`.
- Hyperparameters: max length 128, AdamW, learning rate 2e-5, batch size 16, up to 4 epochs, early stopping (patience 2) on an inner dev split (10% of each training part), class-weighted cross-entropy.
- Models: `aubmindlab/bert-base-arabertv02-twitter`, `UBC-NLP/MARBERT`, `UBC-NLP/MARBERTv2`, `faisalq/SaudiBERT`.
- Tested with Python 3.13, torch 2.11, transformers 4.57.6, scikit-learn 1.6.1 (Google Colab).

AI-use statement

[Group to confirm and complete] AI tools were used for coding support and language editing only. All design decisions and results are the responsibility of the authors.

Team

[Names and university]